

Evaluation of 3D CNN Architecture for Automated Intracranial Aneurysm Detection in CT Angiography

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Abstract

Intracranial aneurysms are localized enlargements of cerebral arteries that often remain asymptomatic but can become life-threatening if undetected. Up to half are diagnosed only after rupture, contributing to approximately 500,000 deaths annually. This study evaluates a deep learning architecture for automated aneurysm detection and localization in medical imaging data. The RSNA Intracranial Aneurysm Detection dataset was converted into 4,235 3D volumes and standardized through voxel spacing normalization, slice thickness adjustment, and resizing. Point annotations were transformed into 3D segmentation masks using spherical regions of interest. A 3D U-Net-based model was trained for voxel-level segmentation, combined with a scan-level detection head to reduce false positives. Training used Dice and cross-entropy loss with data augmentation. Results show a 17.6% improvement in Dice score on positive cases while maintaining low false positive rates. These findings demonstrate the potential of deep learning to support early aneurysm detection and clinical decision-making.

1. Introduction

Cerebral aneurysms are abnormal dilations of intracranial blood vessels that pose a significant risk of rupture, often resulting in subarachnoid hemorrhage, severe neurological injury, or death [5]. Although many aneurysms remain asymptomatic, ruptured events are associated with high morbidity and mortality, making early and accurate detection critical for effective clinical intervention [2,5]. Detecting cerebral aneurysms remains challenging due to their wide variation in size, shape, and anatomical location within complex neurovascular structures.

Currently, cerebral aneurysms are primarily detected using medical imaging modalities such as computed tomography (CT), CT angiography (CTA), and magnetic resonance imaging (MRI), including magnetic resonance angiography (MRA), which is commonly used to visualize the cerebral vasculature [4]. Interpretation of these scans typically relies on expert radiological assessment, which can be time-consuming and subject to inter-observer variability [4]. Small aneurysms or those located near vessel bifurcations are particularly difficult to identify, increasing the risk of missed diagnoses. As medical imaging datasets continue to grow in size and complexity, there is an increasing interest in automated methods that can assist clinicians in identifying aneurysms more efficiently and consistently.

Recent research has explored deep learning methods for automated analysis of volumetric medical images. [10,11]. In particular, convolutional neural networks (CNNs) have demonstrated strong performance in a range of neuroimaging tasks, including vessel segmentation and lesion detection [7]. Three-dimensional CNN architectures such as U-Net variants are especially well-suited for medical imaging because they can capture spatial context in volumetric data while performing precise voxel-level predictions.

Despite these advances, several challenges remain in applying deep learning to cerebral aneurysm detection. Medical imaging datasets often exhibit substantial heterogeneity in spatial resolution, acquisition protocols, and volume dimensions, which can negatively impact model performance. Additionally, many aneurysm datasets provide limited annotations, such as point-level labels rather than full voxel-wise segmentations, complicating supervised training. These factors highlight the need for robust preprocessing pipelines and learning strategies that can operate effectively under weak supervision [3,6].

To address these challenges, this work presents a scalable deep learning pipeline for automated cerebral aneurysm detection and segmentation from multimodal neuroimaging data. Using a dataset of over 4,000 CT, CTA, MRI, and MRA scans [1], we implement standardized preprocessing, weakly supervised mask generation from point annotations, and extensive data augmentation to improve model robustness and generalization. A 3D

convolutional neural network based on a U-Net-style segmentation architecture is used to detect aneurysm regions in volumetric scans. The goal of this research is to evaluate the effectiveness of CNN-based 3D segmentation models for automated intracranial aneurysm detection and to investigate how standardized preprocessing and weakly supervised mask generation can improve model robustness and detection performance.

2. Methods

This study presents a deep learning pipeline for the detection and segmentation of intracranial aneurysms from volumetric neurovascular imaging data. The proposed system consists of four primary stages: dataset preparation, image preprocessing and standardization, region-of-interest extraction, and model training using a three-dimensional convolutional neural network. The overall goal of this pipeline, shown in Figure 1, is to standardize heterogeneous medical imaging data and improve the model’s ability to detect small aneurysm structures within large volumetric scans. After preprocessing, the resulting standardized volumes are used to train a 3D segmentation model based on the U-Net architecture.

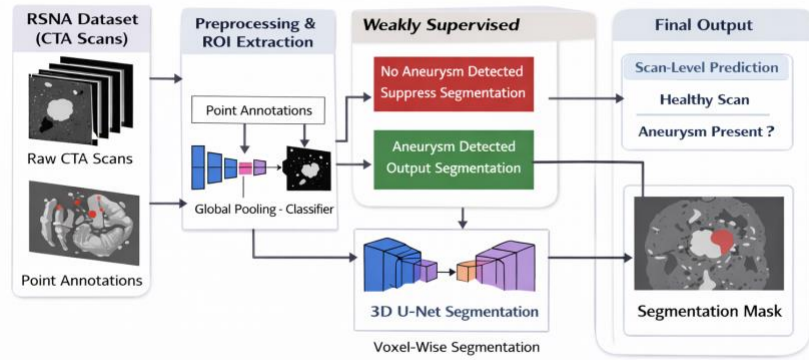


Figure 1: Overview of the proposed aneurysm detection pipeline, including preprocessing, ROI extraction, weakly supervised mask generation, 3D U-Net segmentation, and scan-level classification with false positive suppression.

2.1 Exploratory Data Analysis

The dataset used in this study is derived from the RSNA Intracranial Aneurysm Detection Challenge hosted on the Kaggle Platform [1]. After preprocessing and quality filtering, the dataset used in this work contains 4,235 volumetric neurovascular scans, corresponding to an estimated 4,235 patients. The primary imaging modality is computed tomography angiography (CTA), which provides high-resolution visualization of intracranial blood vessels and is commonly used for aneurysm detection.

Among the 4,235 scans, 2254 scans (46.8%) contain annotated aneurysms, while 1,981 scans (53.2%) represent negative cases without aneurysm findings. Annotations in the dataset are provided as point-level coordinates indicating the location of aneurysms, rather than full voxel-wise segmentation masks. Consequently, additional processing is required to convert these annotations into training labels suitable for volumetric segmentation models.

Exploratory data analysis (EDA) was conducted to better understand the structure of the dataset and identify preprocessing requirements before model training. Initial analysis revealed substantial variability in spatial resolution and image dimensions across imaging series. In particular, voxel spacing and slice thickness varied significantly between scans, reflecting differences in acquisition protocols across imaging centers. Histogram and boxplot visualizations were generated to examine the distribution of image dimensions and voxel spacing across the dataset. These visualizations, shown in Figure 2, highlight the heterogeneity of the raw imaging data and demonstrate the need for standardized spatial resolution and consistent volume dimensions to ensure compatibility with deep learning models and batch-based training procedures.

Demographic analysis was also performed using the available metadata. Within the dataset, 75.5% of aneurysm cases occurred in female patients. The mean age of patients with aneurysms was 60.3 years, compared to 57.1 years for patients without aneurysms. Statistical testing indicated a weak but significant correlation between age and aneurysm presence (Pearson $r = 0.10$, $p < 0.001$), indicating that aneurysm cases in this dataset are more common among older individuals and female patients.

These observations motivated the preprocessing pipeline described in Section 2.2, which standardizes voxel spacing, slice thickness, and volume dimensions to produce consistent volumetric input for model training.

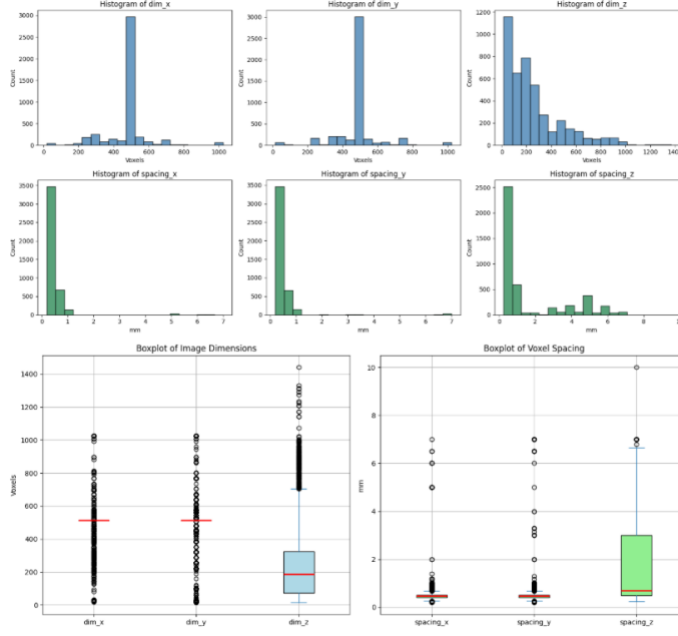


Figure 2: Distribution of image dimensions (dim_x, dim_y, dim_z) and voxel spacing (spacing_x, spacing_y, spacing_z) across the dataset, shown using histograms and boxplots. The visualizations highlight substantial variability in spatial resolution and volume dimensions, motivating the need for preprocessing steps such as voxel spacing normalization and standardized resizing.

2.2 Preprocessing Pipeline

To address inconsistencies in spatial resolution and volume dimensions, a multi-stage preprocessing pipeline was developed to standardize volumetric inputs before model training.

The preprocessing workflow consists of three sequential stages:

- Pixel Spacing Standardization
- Slice Thickness Standardization
- Image Resizing

Each stage operates on the output produced by the previous stage, forming a complete preprocessing pipeline.

2.2.1 Voxel Spacing Standardization

The first stage of preprocessing standardizes the in-plane voxel spacing of each volumetric scan. Due to differences in imaging protocols and scanner configurations across institutions, the raw imaging data exhibit variability in pixel resolution and spatial scale. To ensure consistent anatomical representation across scans, all volumes were resampled to a target in-plane voxel spacing of $0.5 \text{ mm} \times 0.5 \text{ mm}$ in the x and y directions.

This spacing was selected because it preserves sufficient anatomical detail for detecting small vascular structures such as intracranial aneurysms while maintaining manageable computational requirements for deep learning models. Resampling was performed using linear interpolation implemented through the SimpleITK library [10]. The resampling operation adjusts the spatial resolution of each scan to the target spacing while preserving the overall structure of the image.

Standardizing voxel spacing ensures that anatomical features appear at consistent physical scales across the dataset, which improves model stability and facilitates batch-based training of three-dimensional convolutional neural networks.

2.2.2 Slice Thickness Standardization

Following in-plane normalization, slice spacing in the z-dimension was standardized to ensure consistent volumetric representation across scans. Raw neuroimaging studies frequently exhibit varying slice thickness due to differences in reconstruction settings, such as selected slice thickness during image reconstruction and scanner-specific acquisition parameters (e.g., pitch and detector configuration).

To address this variability, all scans were resampled to a target slice thickness of 1.0 mm along the z-axis using linear interpolation. This step produces an approximately isotropic voxel spacing of $0.5 \text{ mm} \times 0.5 \text{ mm} \times 1.0 \text{ mm}$, allowing the model to better capture three-dimensional anatomical structures.

Standardizing slice thickness improves the consistency of volumetric representations and facilitates more effective learning of spatial features within 3D convolutional neural network architectures.

2.2.3 Image Resizing

The final preprocessing stage ensures consistent volume dimensions for neural network input. After voxel spacing and slice thickness standardization, each image volume is adjusted to fixed spatial dimensions of $512 \times 512 \times 256$ voxels. Volumes larger than this

target size are center-cropped, while smaller volumes are zero-padded along each dimension to match the required input shape.

This size was selected to preserve relevant neurovascular structures while remaining computationally feasible for three-dimensional deep learning model training. Standardizing input dimensions ensures compatibility with batch-based processing and consistent memory usage during training.

2.2.4 Region of Interest Cropping

To further improve model performance, region-of-interest (ROI) extraction was applied to reduce irrelevant background information and focus the model on anatomically meaningful regions. ROI extraction reduces the number of background voxels and allows the model to concentrate on regions more likely to contain vascular structures, improving both learning efficiency and detection performance.

Two ROI extraction strategies were implemented to support both annotated and unannotated data.

For Scans with aneurysm annotations, a label-based ROI approach was used. In this method, all non-zero voxels in the annotation masks were identified to compute a tight three-dimensional bounding box surrounding the aneurysm. A fixed spatial margin of 20 voxels was added in each dimension (x, y, z) to include the surrounding vascular context. The resulting bounding box was then applied to both the image volume and its corresponding annotation mask to extract the region used for model training.

For scans without aneurysm annotations, a center-based ROI strategy was applied. In this approach, a fixed-size subvolume centered within the scan was extracted to maintain consistency in input size and anatomical coverage. This ensures that negative samples still contain representative vascular structures while avoiding excessive background regions.

2.3 Two-Stage Detection and Segmentation Framework

The proposed system is a two-stage deep learning pipeline designed for intracranial aneurysm detection in 3D volumetric scans [3]. Architecture integrates voxel-level segmentation with scan-level classification to address the high false positive rates that commonly arise when applying pure segmentation models to medical imaging tasks.

In the first stage, a 3D UNet segmentation network predicts voxel-wise aneurysm probabilities within an ROI centered around the Circle of Willis. In the second stage, a

scan-level detection head evaluates global features derived from the segmentation network and determines whether an aneurysm is present within the scan. If the classifier predicts a healthy scan, all segmentation predictions are suppressed.

This gating mechanism significantly reduces false positives by preventing the segmentation network from producing small spurious detections on healthy scans. The overall architecture, therefore, balances high sensitivity for aneurysm localization with strong specificity for scan-level screening, which is essential for practical clinical deployment.

2.4 Stage 1: 3D UNet Segmentation Network

The segmentation backbone is based on a 3D UNet architecture [11] consisting of an encoder–decoder structure with skip connections between corresponding resolution levels. This design enables the model to capture both global contextual information and fine-grained spatial features, which are necessary for detecting small vascular abnormalities.

The network employs five encoder levels, resulting in four down-sampling operations before reaching the bottleneck. Each level captures features at progressively coarser spatial resolutions, allowing the model to learn fine-grained local structures in early layers and higher-level anatomical context in deeper layers.

The number of feature channels at each encoder level is 32, 64, 128, 256, and 320, respectively. The progressive increase in channel depth compensates for the reduction in spatial resolution at each stage, preserving representational capacity. The bottleneck operates at 320 channels, which was selected as a memory-efficient alternative to higher-capacity configurations (e.g., 512 channels) while maintaining sufficient expressive power for the task.

Down-sampling is performed using stride-2 3D convolutions at each resolution transition, rather than traditional max-pooling. This makes spatial down-sampling a learnable operation, enabling the network to optimize how spatial information is compressed during training. Compared to fixed pooling operations, this approach better preserves task-relevant features.

Each convolutional block consists of 3D convolutional layers followed by instance normalizations and Parametric ReLU (PReLU) activation. PReLU allows the network to learn channel-wise activation slopes for negative inputs, providing greater flexibility than fixed activation functions such as ReLU or Leaky ReLU.

Instance normalization is applied after every convolutional layer. Unlike batch normalization, which relies on batch-level statistics, instance normalization normalizes each sample independently across spatial dimensions. This is particularly advantageous for 3D medical image segmentation, where GPU memory constraints limit training to **small batch sizes (batch size = 2)**, making batch statistics unreliable.

The decoder path mirrors the encoder structure, using upsampling followed by skip connections to recover spatial resolution and refine localization. A final **1×1×1 convolution** with **sigmoid activation** produces a voxel-wise probability map representing the likelihood of aneurysm presence at each voxel.

Input volumes consist of $128 \times 128 \times 128$ voxel patches extracted from the ROI centered on the Circle of Willis. Before input into the network, Hounsfield Unit (HU) intensity windowing is applied using a range of $[-200, 400]$ to suppress irrelevant anatomical structures and emphasize vascular regions.

The network produces a voxel-wise probability map representing the likelihood of aneurysm presence at each voxel.

2.4.1 Segmentation Loss

To address the extreme class imbalance between aneurysm voxels and background tissue, the segmentation loss combines **binary cross-entropy (CE)** with **Dice loss**:

$$L_{\text{seg}} = \lambda_{\text{CE}} \cdot \text{CE}(\text{pred}, \text{target}) + \text{DiceLoss}(\text{pred}, \text{target})$$

where $\lambda_{\text{CE}} = 3.0$ places additional weight on the cross-entropy term to stabilize optimization under severe foreground sparsity.

2.5 Stage 2: Scan-Level Detection Head

While voxel-wise segmentation provides detailed localization, segmentation networks may produce small false positive regions in scans without aneurysms. To mitigate this issue, a scan-level detection head is incorporated into the architecture.

The detection head aggregates high-level features from the UNet output using a global pooling operation, followed by a fully connected layer that produces a binary classification score indicating the presence or absence of an aneurysm within the scan.

During inference, this classification output is used as a gating mechanism to refine the segmentation results. Specifically, if the predicted probability of aneurysm presence falls below a threshold of 0.5, the scan is classified as negative, and the corresponding

segmentation output is suppressed. Conversely, if the predicted probability exceeds the threshold, the scan is classified as positive, and the segmentation mask is retained.

This design reduces false-positive detections by preventing the model from reporting small, noisy segmentation regions in scans that are predicted to be free of pathology.

2.6 Joint Training Objective

The segmentation network and detection head are trained jointly using a multi-task objective:

$$L_{total} = L_{seg} + w_{det} \cdot L_{det}$$

where:

- L_{seg} — segmentation loss
- L_{det} — binary classification loss
- w_{det} — weighting factor controlling the influence of the detection task.

During training, the detection weight is gradually reduced to shift focus on segmentation refinement as training progresses.

2.7 Model Selection Criterion

To select the optimal checkpoint during training, a composite validation metric is used:

$$Score = Dice_{POS} \times (1 - FP_{rate})$$

This metric balances two critical objectives:

- **Sensitivity** — accurate segmentation on aneurysm cases
- **Specificity** — low false positive rates on healthy scans

A model that achieves high Dice scores by over-predicting aneurysm regions would incur a penalty through an increased false positive rate.

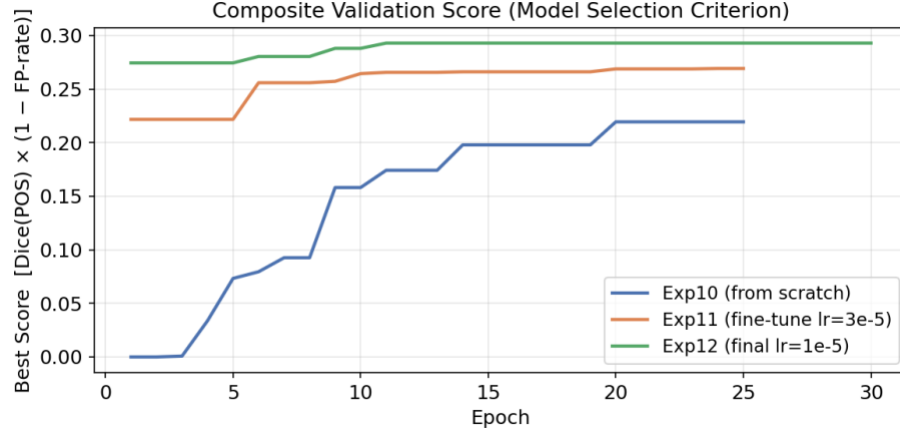


Figure 3: Composite validation score across training epochs for the three training stages. The progressive fine-tuning strategy allows later stages to initialize from stronger checkpoints and achieve higher final validation performance.

2.8 Progressive Fine-Tuning Strategy

Training is performed using a three-stage progressive fine-tuning schedule, where each stage initializes from the best checkpoint of the previous experiment and uses a reduced learning rate.

Stage	GPU	Learning Rate	Det weight	Epochs
Exp10 (baseline)	V100 32GB	$1e^{-4}$	2.0	25
Exp11 (fine-tune)	V100 32GB	$3e^{-5}$	1.5	25
Exp12 (final)	H100 80GB	$1e^{-5}$	1.0	30

Table 1: Training configuration for each experimental stage, including GPU type, learning rate, detection loss weight, and number of training epochs used in the progressive fine-tuning strategy.

This strategy avoids catastrophic forgetting while enabling the network to gradually refine feature representations.

Additional training parameters include:

- batch size = 2
- gradient accumulation = 4 (effective batch size 8)

- mixed precision training (AMP)
- positive case oversampling ratio = 1.0
- minimum detection cluster size = 50 voxels
- cosine annealing learning rate scheduler.

3.0 Results

3.1 Training Dynamics

The training loss across the three training stages is shown in Figure 4.

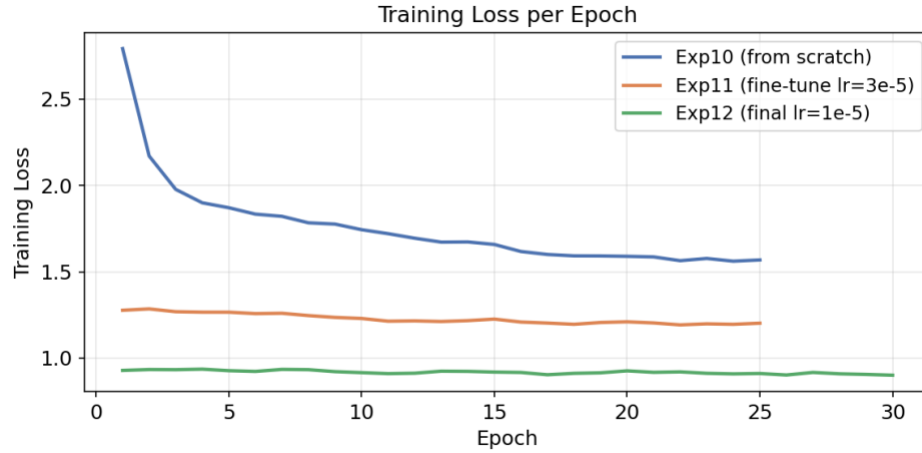


Figure 4: Training and validation loss across epochs for the three training stages (Exp10, Exp11, Exp12), showing improved convergence during progressive fine-tuning.

The baseline experiment (Exp10) exhibits a steep reduction in loss during early epochs as the model learns from randomly initialized weights. In contrast, the fine-tuning stages (Exp11 and Exp12) begin with substantially lower loss values, reflecting the benefit of initializing them from previously trained checkpoints.

Exp12 achieves the lowest final loss, indicating improved convergence and model refinement during the final training stage.

3.2 Validation Performance

Validation Dice scores across training epochs are presented in Figure 5.

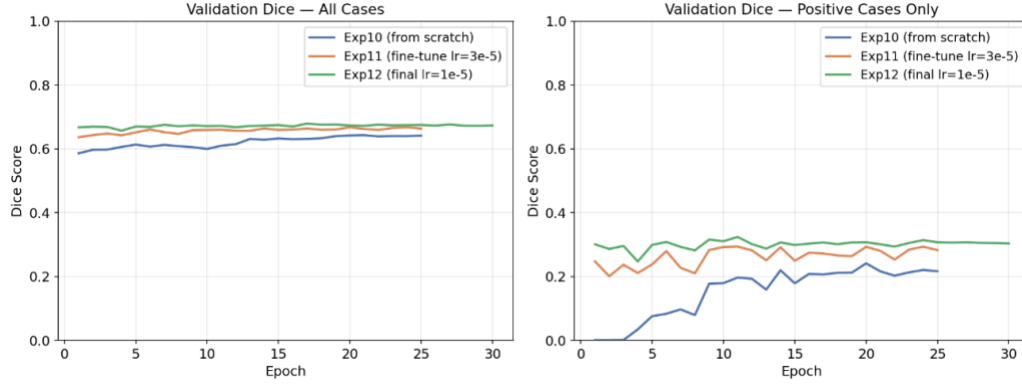


Figure 5: Validation Dice scores across training epochs, including Dice over all scans and Dice restricted to positive aneurysm cases, demonstrating improved segmentation performance in later training stages.

Two metrics are shown:

- Dice across all scans
- Dice restricted to positive aneurysm cases.

Dice scores across all scans remain relatively stable due to the large number of healthy scans in the dataset. However, Dice performance on positive cases demonstrates a clear upward trend across the progressive training stages, with Exp12 consistently outperforming earlier models.

3.3 False Positive Analysis

The false positive rate on negative scans across training epochs is shown in Figure 6.

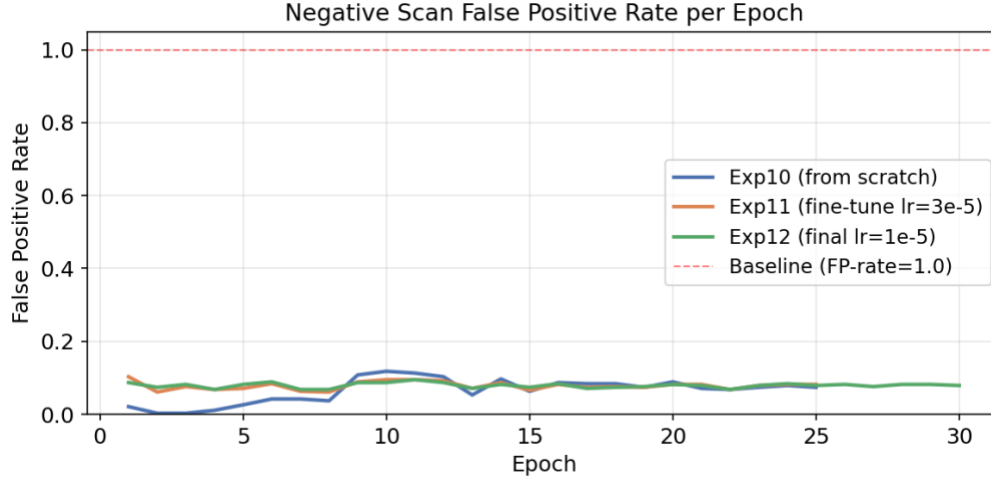


Figure 6: False positive rate on negative scans across training epochs for all experiments, showing stable and low false positive behavior due to the scan-level detection head.

All experiments rapidly converge to stable false positive rates between 5–10%, which is substantially lower than the random baseline (FP-rate = 1.0). The detection head plays a key role in maintaining low false positive rates across all stages.

3.4 Quantitative Test Set Performance

Final performance metrics on the held-out test set are summarized in Table 2 and Figure 7.

Model	Dice (All)	Dice (POS)	FP-rate	Composite Score
Exp10	0.6609	0.2921	0.112	0.220
Exp11	0.6745	0.3153	0.104	0.270
Exp12	0.6789	0.3435	0.115	0.295

Table 2: Quantitative performance metrics on the test set for each training stage, including Dice score (all scans), Dice scores on positive cases, false positive rate, and composite score.

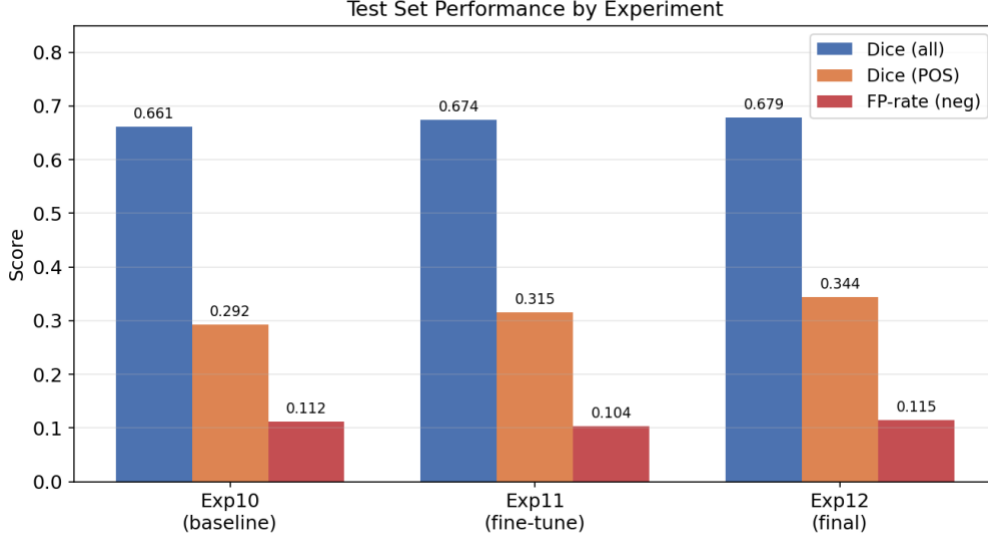


Figure 7: Summary of model performance on the test set, illustrating improvements in Dice score and composite metric across progressive training stages.

The progressive fine-tuning strategy yields consistent improvements across training stages. The final model (Exp12) achieves the highest segmentation accuracy on positive scans while maintaining a low false positive rate.

Notably, Dice (POS) improves by 17.6% relative to the baseline ($0.2921 \rightarrow 0.3435$).

4.0 Discussion

The experimental results demonstrate that the proposed two-stage detection and segmentation pipeline effectively balances sensitivity and specificity, which is essential for clinical screening applications [9, 10]. By combining voxel-wise segmentation with scan-level classification, the model achieves accurate localization while reducing the likelihood of false-positive detections.

The inclusion of a scan-level detection head plays a critical role in improving overall model reliability. Standalone segmentation models are prone to generating small false-positive regions in healthy scans; however, the proposed gating mechanism suppresses these spurious predictions when no aneurysm is detected at the scan level. This results in a false-positive rate below 12% across all experiments, indicating a meaningful improvement in specificity.

Another key finding is the effectiveness of the progressive fine-tuning strategy. Rather than training a single model for an extended number of epochs, the staged training

approach enables the network to iteratively refine and reduce learning rates, leading to more stable convergence and consistent improvements in Dice(POS) across training stages.

It is important to note that the Dice scores computed across all scans can be influenced by the proportion of negative cases in the dataset. Because negative scans contain no positive voxels, they may artificially inflate overall performance metrics. As a result, Dice (POS) provides a more meaningful and relevant evaluation of segmentation performance for aneurysm detection.

Compared to prior work, the proposed approach demonstrates improved robustness in handling false-positive predictions, a common limitation in aneurysm detection models. Traditional segmentation-based methods often rely solely on voxel-wise predictions, which can lead to over-segmentation in healthy scans. By incorporating a scan-level classification component, this work introduces an additional level of decision-making that improves overall detection reliability. This aligns with recent trends in medical imaging research that emphasize hybrid detection-segmentation frameworks for improved performance [9, 10].

From a clinical perspective, reducing false-positive detections is particularly important in neurovascular imaging workflows. Excessive false positives can increase radiologist workload and lead to unnecessary follow-up imaging or interventions. The proposed pipeline, by improving specificity while maintaining sensitivity, has the potential to function as a decision-support tool that assists radiologists in identifying aneurysms more efficiently and consistently. Such systems could be integrated into clinical workflows to prioritize high-risk scans and reduce diagnostic burden.

Despite these improvements, several limitations should be considered. First, the model relies on **point-level annotations** that are converted into approximate segmentation masks, which may not accurately capture the true shape and extent of aneurysms. Second, although preprocessing standardization improves consistency, variability in imaging quality and scanner characteristics may still impact model performance. Third, the use of fixed input resolution may limit the model’s ability to detect very small aneurysms. Finally, while the dataset is relatively large, it may not fully represent all patient populations or rare aneurysm types, which could affect generalizability.

Future work may address these limitations by incorporating higher-resolution inputs, more precise voxel-wise annotations, and advanced architectures, such as multi-scale or attention-based models, to improve localization accuracy [11].

5.0 Conclusion

This work presents a two-stage deep learning framework for intracranial aneurysm detection in volumetric medical imaging [3]. The proposed system integrates a 3D UNet segmentation network with a scan-level detection head that suppresses false positives on healthy scans.

Experimental results demonstrate that the approach achieves improved segmentation accuracy on aneurysm cases while maintaining low false positive rates, which is critical for real-world clinical deployment [10]. The progressive fine-tuning training strategy further enhances model performance by enabling stable convergence and iterative refinement of learned features.

Overall, the proposed method provides a promising direction for automated aneurysm detection systems that require both high sensitivity for pathology localization and strong specificity for clinical reliability [2].

References

1. RSNA. RSNA Intracranial Aneurysm Detection Challenge Dataset [Internet]. Kaggle; 2023. Available from: <https://www.kaggle.com/competitions/rsna-intracranial-aneurysm-detection>
2. Brain Aneurysm Foundation. Brain aneurysm statistics and facts [Internet]. Brain Aneurysm Foundation; 2024. Available from: <https://www.bafound.org/understanding-brain-aneurysms/statistics-and-facts/>
3. Rainville P, et al. Weakly supervised intracranial aneurysm detection and segmentation in MR angiography. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*; 2025. Available from: https://openaccess.thecvf.com/content/ICCV2025W/CVAMD/papers/Rainville_Weakly_Supervised_Intracranial_Aneurysm_Detection_and_Segmentation_in_MR_angiography_ICCVW_2025_paper.pdf
4. Schaafsma JD, Velthuis BK, Majoie CBLM, et al. A prospective trial of 3T and 1.5T time-of-flight and contrast-enhanced MR angiography in the follow-up of coiled intracranial aneurysms. *American Journal of Neuroradiology*. 2008;29(1):111–116.
5. Etminan N, Rinkel GJE. Management of aneurysmal subarachnoid hemorrhage: state of the art and future perspectives. *Surgical Neurology International*. 2016;7(Suppl 1):S13–S18. Available from: <https://surgicalneurologyint.com/surgicalint-articles/management-of-aneurysmal-subarachnoid-hemorrhage-state-of-the-art-and-future-perspectives/>

6. Rasoulilian A, Salari S, Xiao Y. Weakly supervised segmentation of intracranial aneurysms using a novel 3D focal modulation U-Net [Internet]. arXiv; 2023. Available from: <https://arxiv.org/>
7. Iqbal MS, Belal Bin Heyat M, Parveen S, Ammar Bin Hayat M, Roshanzamir M, Alizadehsani R, Akhtar F, Sayeed E, Hussain S, Hussein HS, Sawan M. Progress and trends in neurological disorders research based on deep learning. *Comput Med Imaging Graph*. 2024 Sep;116:102400. doi:10.1016/j.compmedimag.2024.102400.
8. Hilbert A, et al. BRAVE-NET: fully automated arterial brain vessel segmentation in patients with cerebrovascular disease. *Front Artif Intell*. 2020;3:552258.
9. Bayram F, et al. A systematic review of deep learning in MRI-based cerebrovascular disease diagnosis. *J Neuroradiol*. 2025.
10. Litjens G, Kooi T, Bejnordi BE, et al. A survey on deep learning in medical image analysis. *Med Image Anal*. 2017;42:60–88.
11. Ronneberger O, Fischer P, Brox T. U-Net: convolutional networks for biomedical image segmentation. In: *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*; 2015. p. 234–241.